STANDARD COLLEGE NTUNGAMO

S6 2026 TEST 1

Time: 1 hour

1. At a construction site, engineers are installing copper and steel wires for supporting equipment and electrical connections in a new laboratory building. During testing, one copper wire is pulled until it eventually breaks so that the engineers can study how the material behaves under tension. Later, they also observe that a steel wire fixed between two rigid supports changes length when the temperature drops during the night. During the experiment, the engineers record how the copper wire behaves as the pulling force increases.

During a cold night, a steel wire of cross-sectional area $1mm^2$ fixed between two walls cools from 40°C to 20°C.

(Take Young's modulus of steel $=2.0\times10^{11}$ Pa and coefficient of linear expansion $=1.1\times10^{-5}\,K^{-1}$)

TASK

Using your knowledge of material properties, answer the following questions:

(a) Explain the meaning of the following terms as used in testing materials:

i. Tensile stress
ii. Tensile strain

(b) i). Sketch the stress–strain graph for the copper wire and explain the main features of the graph.
ii). Describe what happens to the energy supplied to stretch the wire at the different stages of the graph.
iii). Derive the expression for the work done to stretch the copper wire by a distance, e, if its force constant is K.

(**c**) i.) Calculate the strain produced in the wire due to the change in temperature.
ii.)Determine the force required to prevent the wire from contracting.

(**d**) Explain what engineers mean by work hardening and why it is important when designing metal components.

2. At a children's amusement park, a small toy ball attached to a string is swung in a horizontal circular path during a physics demonstration to explain circular motion.
A ball of mass 0.15 kg is tied to a string and whirled so that it moves in a circle of radius 1.82 m in a horizontal plane. The ball completes 10 revolutions in 18 seconds while moving at a steady speed.

Using this information, determine:

i. The speed of the ball.
ii. The centripetal force acting on the ball.
iii. The tension in the string.

END